Alzheimer's Disease Classification using Biomarker-Guided Attention Network (BGAN)

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***Abstract*—Alzheimer's disease (AD) is a progressive neurodegenerative disorder responsible for 60–80% of all dementia cases worldwide, affecting over 55 million individuals. Early and accurate staging of AD severity from structural Magnetic Resonance Imaging (MRI) is clinically critical, as neuroanatomical changes precede observable symptoms by years. Existing deep learning classifiers achieve high accuracy but operate as black boxes, making them unsuitable for direct clinical deployment. In this paper, we propose the Biomarker-Guided Attention Network (BGAN), a novel architecture that integrates a DenseNet121 backbone with a dual-attention mechanism — comprising Squeeze-and-Excitation channel attention and CBAM-style spatial attention — guided by a Gaussian biomarker prior mask encoding nine clinically established neuroanatomical regions. A learnable blend parameter $\hat{\alpha}$ balances data-driven and prior-guided spatial attention during end-to-end training. A Composite Loss combining Cross-Entropy and a λ-warmed Biomarker Alignment Loss ($\mathcal{L}_{bio}$) directly enforces attention concentration in biomarker regions during training. We further propose the Biomarker Alignment Score (BAS), a novel quantitative interpretability metric that measures the ratio of attention mass within known biomarker regions to that outside. Experiments on a 44,000-image four-class Alzheimer's MRI dataset with group-level stratified splitting to prevent augmented near-duplicate leakage show that BGAN achieves a BAS of 11.585 $\pm$ 0.482 — more than 8.5$\times$ the best baseline — while maintaining competitive classification performance (Accuracy: 90.81%, F1-Macro: 0.9185, ROC-AUC: 0.9861). Notably, VGG16 achieves 94.14% accuracy with a BAS of 0.000, demonstrating that high accuracy does not imply clinically meaningful attention. BGAN provides the only model that is simultaneously competitive in accuracy and interpretable in its spatial attention behavior.**

***Index Terms*—Alzheimer's Disease, MRI Classification, Attention Mechanism, Biomarker-Guided Learning, Interpretability, GradCAM, DenseNet121, Transfer Learning, Biomarker Alignment Score**

I. Introduction

Alzheimer's disease (AD) has emerged as one of the most pressing public health concerns of the modern era, driven by increasing life expectancy and aging populations worldwide. It is the most common cause of dementia, characterized by progressive cognitive decline, memory impairment, and loss of functional independence. Recent global estimates indicate that tens of millions of individuals are currently living with dementia, with a rapidly growing incidence rate each year. In countries like India, the burden is particularly significant and is projected to rise sharply in the coming decades due to demographic transitions. Despite extensive advancements in neuroscience and clinical research, no definitive cure exists for Alzheimer's disease, making early diagnosis and accurate staging crucial for delaying progression and improving patient care outcomes.

Neuroimaging plays a central role in the diagnosis and monitoring of Alzheimer's disease. Among available modalities, structural Magnetic Resonance Imaging (MRI) is widely adopted due to its non-invasive nature and its ability to capture detailed anatomical information. Characteristic changes such as hippocampal atrophy, ventricular expansion, and cortical thinning in regions like the entorhinal cortex serve as early indicators of disease onset, often appearing well before the manifestation of clinical symptoms. However, manual interpretation of MRI scans by radiologists is inherently challenging. It is not only time-intensive but also subject to inter-observer variability, particularly when distinguishing between adjacent disease stages where structural differences are subtle and difficult to quantify visually.

The rise of deep learning has significantly transformed medical image analysis, offering automated and scalable solutions for disease detection. Convolutional Neural Networks (CNNs), especially when combined with transfer learning from large-scale datasets, have demonstrated strong performance in classifying Alzheimer's stages using MRI data. Architectures such as DenseNet, ResNet, and VGG have been extensively explored, achieving high classification accuracy across multiple benchmarks. Nevertheless, these models often function as opaque systems, lacking transparency in their decision-making processes. More importantly, they may rely on dataset-specific artifacts or irrelevant features, leading to spurious correlations that hinder generalization and reduce clinical reliability.

To improve interpretability, post-hoc explanation techniques such as Grad-CAM have been widely employed to visualize model attention. While these methods provide some insight into model behavior, they are not inherently constrained to align with medically meaningful regions. As a result, attention maps generated by such approaches frequently highlight areas that do not correspond to known neuroanatomical biomarkers of Alzheimer's disease. This misalignment raises concerns regarding the trustworthiness of deep learning models in

clinical applications, where interpretability is as important as predictive performance.

A promising direction to address this limitation involves integrating domain-specific knowledge directly into the learning process. In the context of Alzheimer's disease, decades of clinical research have identified specific brain regions that are consistently affected during disease progression. Incorporating these neuroanatomical priors into deep learning frameworks can guide models toward more meaningful feature representations. However, effectively combining data-driven learning with prior knowledge in a flexible and adaptive manner remains a challenging research problem.

In this work, we propose a novel Biomarker-Guided Attention Network (BGAN) that bridges this gap by explicitly aligning model attention with established neuroanatomical biomarkers. The proposed architecture is built upon a DenseNet121 backbone and incorporates both channel-wise and spatial attention mechanisms inspired by squeeze-and-excitation (SE) and convolutional block attention module (CBAM) designs. A key innovation of our approach is the integration of a precomputed Gaussian biomarker prior mask, representing clinically relevant brain regions associated with Alzheimer's pathology. This prior is combined with the learned spatial attention map through a trainable blending parameter, allowing the model to dynamically balance between data-driven features and domain knowledge.

To further enforce biologically meaningful learning, we introduce a composite training objective that extends beyond standard classification loss. In addition to cross-entropy loss with label smoothing, we incorporate a Biomarker Alignment Loss that penalizes attention assigned outside predefined biomarker regions.

Moreover, we propose a novel evaluation metric, termed the Biomarker Alignment Score (BAS), designed to quantitatively assess how well a model's attention aligns with clinically significant regions. Unlike conventional performance metrics that focus solely on prediction accuracy, BAS provides an architecture-agnostic measure of interpretability, enabling systematic comparison across different models and attention mechanisms.

The primary contributions of this work are threefold. First, we introduce a biomarker-guided attention framework that integrates neuroanatomical priors into deep learning for Alzheimer's disease classification. Second, we design a learnable attention blending mechanism that adaptively combines spatial attention with biomarker information. Third, we develop both a training objective and an evaluation metric that explicitly promote and quantify biologically meaningful attention.

By jointly optimizing for accuracy and interpretability, the proposed BGAN framework advances the development of trustworthy AI systems in medical imaging. It offers a step toward clinically viable solutions that not only perform well but also provide explanations aligned with established medical knowledge, thereby enhancing confidence among healthcare practitioners.

II. Related Work

A. CNN-Based AD Classification

Deep learning has become a cornerstone in automated Alzheimer's disease (AD) diagnosis, particularly through the use of Convolutional Neural Networks (CNNs) applied to structural MRI data. Due to the limited size of medical imaging datasets, transfer learning from ImageNet-pretrained models has emerged as a widely adopted strategy. These pretrained architectures enable efficient feature extraction while reducing the need for large-scale labeled medical datasets.

Early studies leveraged classical CNN architectures such as VGG16, ResNet, and DenseNet for binary and multi-class AD classification. Nawaz *et al.* [5] demonstrated that fine-tuning VGG16 on MRI slices can yield strong performance, achieving over 92% accuracy on publicly available datasets. Similarly, Zhang *et al.* [6] explored ensemble strategies using ResNet-50, showing that combining multiple models can improve robustness and push classification accuracy beyond 94% on the ADNI dataset.

DenseNet-based architectures have gained particular attention due to their dense connectivity pattern, which promotes feature reuse and mitigates vanishing gradient issues. Originally proposed by Huang *et al.* [7] for general image recognition tasks, DenseNet121 has been successfully adapted for medical imaging applications. Farooq *et al.* [8] demonstrated its effectiveness for multi-class AD staging, highlighting its parameter efficiency and strong generalization capabilities.

Despite these advancements, most CNN-based approaches primarily focus on improving classification accuracy, often overlooking interpretability and clinical relevance. These models may inadvertently learn dataset-specific artifacts or background patterns that are not related to disease pathology. As a result, their predictions, while accurate, may lack reliability when deployed in real-world clinical settings.

B. Attention Mechanisms in Medical Imaging

To enhance feature representation and improve model focus on relevant regions, attention mechanisms have been widely incorporated into deep learning architectures. Channel attention mechanisms, such as the Squeeze-and-Excitation (SE) block introduced by Hu *et al.* [9], enable networks to adaptively recalibrate feature channels by modeling interdependencies between them. This allows the network to emphasize informative features while suppressing less relevant ones.

Building upon this idea, Woo *et al.* [10] proposed the Convolutional Block Attention Module (CBAM), which combines both channel and spatial attention. The spatial attention component generates a 2D attention map that highlights important regions in the input feature map, providing additional localization capability. These attention modules have been successfully integrated into various medical imaging tasks, including disease classification and segmentation, leading to consistent performance improvements.

In neuroimaging applications, attention-based models have shown promise in identifying disease-relevant regions. For

instance, Schlemper *et al.* [12] introduced attention gating mechanisms for brain tumor segmentation, demonstrating improved localization and interpretability. However, existing attention mechanisms are typically learned in a purely data-driven manner, without incorporating prior medical knowledge. Consequently, the learned attention maps may not align with clinically validated anatomical regions associated with Alzheimer's disease.

C. Interpretability and Explainability

Interpretability is a critical requirement for deploying deep learning models in clinical environments. Post-hoc explanation techniques, such as Grad-CAM [11], have become widely used for visualizing model decisions. Grad-CAM generates class-specific activation maps by leveraging gradient information from convolutional layers, providing insights into which regions influence the model's predictions.

Despite its popularity, Grad-CAM has several limitations. It is not part of the training process and therefore does not influence how the model learns features. Additionally, its outputs are sensitive to architectural choices, such as the selection of the target convolutional layer, and may vary significantly across models. More importantly, Grad-CAM does not guarantee that highlighted regions correspond to meaningful anatomical structures, which limits its reliability in medical applications.

Recent research has explored incorporating interpretability directly into the training process through attention regularization and weak supervision. For example, Singh *et al.* [13] proposed attention-guided learning frameworks for medical imaging tasks, encouraging models to focus on relevant regions. However, such approaches are often task-specific and lack a standardized method for evaluating the alignment between model attention and known disease biomarkers.

In the context of Alzheimer's disease, there remains a significant gap in both enforcing and quantitatively measuring the alignment of model attention with neuroanatomical biomarkers. Existing works primarily rely on qualitative visualization, which is subjective and difficult to compare across models. To address this limitation, our work introduces the Biomarker Alignment Score (BAS), a novel metric designed to objectively quantify how well a model's attention corresponds to clinically relevant brain regions. This provides a standardized framework for evaluating interpretability alongside classification performance.

III. Methodology

A. Dataset and Preprocessing

We use the Alzheimer's MRI Multiclass Dataset [37] from Kaggle, comprising 44,000 labeled 2D axial brain MRI images across four classes: *NonDemented* (12,800), *VeryMildDemented* (11,200), *MildDemented* (10,000), and *ModerateDemented* (10,000). All images are RGB JPEG files resized to 224×224.

A key challenge with this dataset is that it is pre-augmented, meaning multiple variants of the same source scan exist. A standard random image-level split would allow augmented twins to appear in both training and test sets, inflating evaluation metrics. We address this with a three-stage preprocessing pipeline:

Stage 1 — MD5 Deduplication: All images are MD5-hashed. Exact byte-identical duplicates (462 found) are removed, leaving 43,538 unique images.

Stage 2 — Perceptual Hash Clustering: A perceptual hash (pHash) is computed for each image at 16×16 resolution. Image pairs within the same class with Hamming distance ≤ 6 are treated as near-duplicates and grouped using Union-Find, identifying 29,023 near-duplicate pairs and 27,273 unique source groups.

Stage 3 — Group-Level Stratified Split: Splits are performed at the group (source scan) level, ensuring all augmented variants of a given scan belong to the same partition. The resulting split is 32,341 train / 5,683 val / 5,514 test, with zero cross-split hash overlap verified by assertion.

Training data augmentation includes random horizontal flip ($p = 0.5$), rotation ($\pm15°$), random resized crop (scale 0.85–1.0), color jitter (brightness/contrast ±0.2), and affine translation ($\pm5\%$). Validation and test data use only resize and ImageNet normalization. *Robustness to Augmentation Bias:* The presence of synthetic augmentations in the dataset introduces a risk of overfitting to transformation-specific patterns rather than underlying anatomical structures. By clustering perceptually similar images and enforcing group-wise splitting, the proposed pipeline ensures that the model learns invariant features rather than memorizing augmented variants.

Generalization Considerations: The preprocessing strategy is specifically designed to simulate real-world deployment conditions, where unseen patient scans may differ significantly from training data. By eliminating duplicate leakage and enforcing strict partitioning, the resulting evaluation metrics more accurately reflect real-world generalization performance.

Input Standardization: All MRI slices are resized to a fixed spatial resolution of 224×224 to match the input requirements of pretrained CNN architectures. This uniformity ensures consistent feature extraction while preserving key anatomical structures necessary for classification.

B. BGAN Architecture

The proposed Biomarker-Guided Attention Network (BGAN) is shown in Fig. 1. It is built on top of a DenseNet121 backbone and combines standard deep feature extraction with attention-based refinement and anatomical prior guidance. The main objective of this design is to retain the strong representational capacity of pretrained CNN features while improving interpretability through biomarker-aware attention.

DenseNet121 is used as the backbone because it provides strong feature reuse, efficient gradient flow, and good transfer learning behavior for medical image analysis. The backbone extracts a deep feature tensor from the input MRI slice, which is then refined using channel and spatial attention before classification. In this way, the model can focus on the most

informative regions of the brain instead of relying on global features alone.

Channel Attention (SE-style): Following [9], the channel attention module computes a weight vector $\mathbf{w}_{ch} \in \mathbb{R}^C$ that captures the relative importance of each feature channel:

$$\mathbf{w}_{ch} = \sigma\Big(\mathrm{FC}_2(\mathrm{ReLU}(\mathrm{FC}_1(\mathrm{GAP}(\mathbf{F})))) + \mathrm{FC}_2(\mathrm{ReLU}(\mathrm{FC}_1(\mathrm{GMP}(\mathbf{F}))))\Big) \tag{1}$$

where $\mathbf{F} \in \mathbb{R}^{B\times C\times H\times W}$ is the feature map and the bottleneck reduction ratio is $r = 16$. This module helps the network emphasize channels that carry more discriminative information for classification while suppressing less useful responses.

Spatial Attention (CBAM-style): Following [10], the spatial attention module produces a spatial map $A_{sp} \in \mathbb{R}^{B\times 1\times H\times W}$:

$$A_{sp} = \sigma(\mathrm{Conv}_{7\times 7}([\mathrm{AvgPool}_C(\mathbf{F}') \,\|\, \mathrm{MaxPool}_C(\mathbf{F}')])) \tag{2}$$

where $\mathbf{F}' = \mathbf{F} \cdot \mathbf{w}_{ch}$. This map highlights spatial regions that are most relevant after channel-wise refinement, allowing the model to focus on informative anatomical locations in the MRI slice.

Biomarker Prior Mask: The prior mask $M \in [0,1]^{H\times W}$ encodes nine Alzheimer's disease-relevant neuroanatomical regions as a weighted Gaussian mixture:

$$M(y,x) = \frac{1}{Z}\sum_{k=1}^{9} w_k \exp\left(-\frac{1}{2}\left[\frac{(y/H - c_y^{(k)})^2}{\sigma_y^{(k)2}} + \frac{(x/W - c_x^{(k)})^2}{\sigma_x^{(k)2}}\right]\right) \tag{3}$$

The nine regions encoded include the bilateral hippocampus, entorhinal cortex, temporal lobe, lateral ventricle, and posterior parietal cortex with region-specific weights. A soft Gaussian prior is used instead of a hard binary mask because MRI anatomy varies across subjects, and a smooth prior allows the model to remain flexible while still receiving anatomical guidance.

Learnable Blend:

$$A^* = \hat{\alpha} \cdot A_{sp} + (1-\hat{\alpha}) \cdot M_{bio} \tag{4}$$

where $\hat{\alpha} = \sigma(\alpha)$ and $\alpha \in \mathbb{R}$ is a learnable scalar initialized to 0. This blending mechanism allows the model to dynamically balance learned spatial attention against the biomarker prior during training. The attended feature map is then computed as $\tilde{\mathbf{F}} = \mathbf{F}' \cdot A^*$.

C. Composite Loss Function

The total training objective combines classification accuracy with biomarker alignment:

$$\mathcal{L} = \mathcal{L}_{CE}(\hat{y}, y) + \lambda(t) \cdot \mathcal{L}_{bio}(A^*) \tag{5}$$

$$\mathcal{L}_{bio}(A^*) = 1 - \frac{\mathbb{E}[A^* \odot M_{bio}]}{\mathbb{E}[A^*] + \varepsilon} \tag{6}$$

$$\lambda(t) = \lambda_0 + \frac{t}{T_w}(\lambda_1 - \lambda_0), \quad t \le T_w \tag{7}$$

where $\mathcal{L}_{CE}$ uses label smoothing with $\epsilon = 0.05$, $\lambda_0 = 0.02$, $\lambda_1 = 0.1$, and $T_w = 8$ epochs. The classification term ensures correct prediction of the Alzheimer's class, while the biomarker alignment term encourages the model to attend to clinically meaningful regions. The warmup schedule prevents the alignment loss from dominating too early in training, when the backbone still needs to learn basic discriminative features.

D. Biomarker Alignment Score (BAS)

To quantitatively measure interpretability, we compute the Biomarker Alignment Score (BAS):

$$\mathrm{BAS} = \frac{\mathrm{mean}(A^*[M_{bio} > \theta])}{\mathrm{mean}(A^*[M_{bio} \le \theta])} \tag{8}$$

where $\theta = 0.3$ thresholds the mask into biomarker and non-biomarker regions. A BAS value greater than 1 indicates that the model attends more strongly to biomarker regions than to the rest of the image. For baseline models, A^* is replaced by Grad-CAM maps upsampled to 224×224. Since BAS is computed independently of training, it serves as an architecture-agnostic measure of attention alignment.

E. Training Protocol

All models follow an identical two-phase protocol. In **Phase 1** (4 epochs, learning rate $= 10^{-3}$), the backbone is frozen and only the classification head is trained. For BGAN, the attention modules are also trained in this phase. This helps the newly added layers adapt quickly to the target dataset without disturbing pretrained features.

In **Phase 2** (10 epochs, learning rate $= 10^{-4}$), the last backbone block is unfrozen and fine-tuned. Specifically, DenseNet121 uses DenseBlock4 and norm5, ResNet50 uses Layer4, and VGG16 uses the last four convolutional layers. Fine-tuning is performed using AdamW with weight decay $\lambda_{wd} = 10^{-4}$ and a cosine annealing learning-rate schedule. Early stopping with patience 4 is applied based on validation accuracy to reduce overfitting.

All experiments are executed with Automatic Mixed Precision (AMP) on an NVIDIA Tesla T4 GPU with 15.6 GB VRAM, using batch size 64. AMP reduces memory usage and speeds up training while maintaining numerical stability, which makes the overall pipeline more efficient for long experiments.

IV. Experiments and Results

A. Implementation Details

The full pipeline is implemented in PyTorch 2.x on Google Colab. Table I summarizes model sizes. BGAN adds only ≈133K parameters over DenseNet121, mainly due to the attention modules and the learnable scalar α, which results in negligible computational overhead compared with the baseline backbone. This makes BGAN a practical extension rather than a heavy architectural redesign. Training time per model is approximately 51–54 minutes on a T4 GPU, which is suitable for rapid experimentation and iterative development.

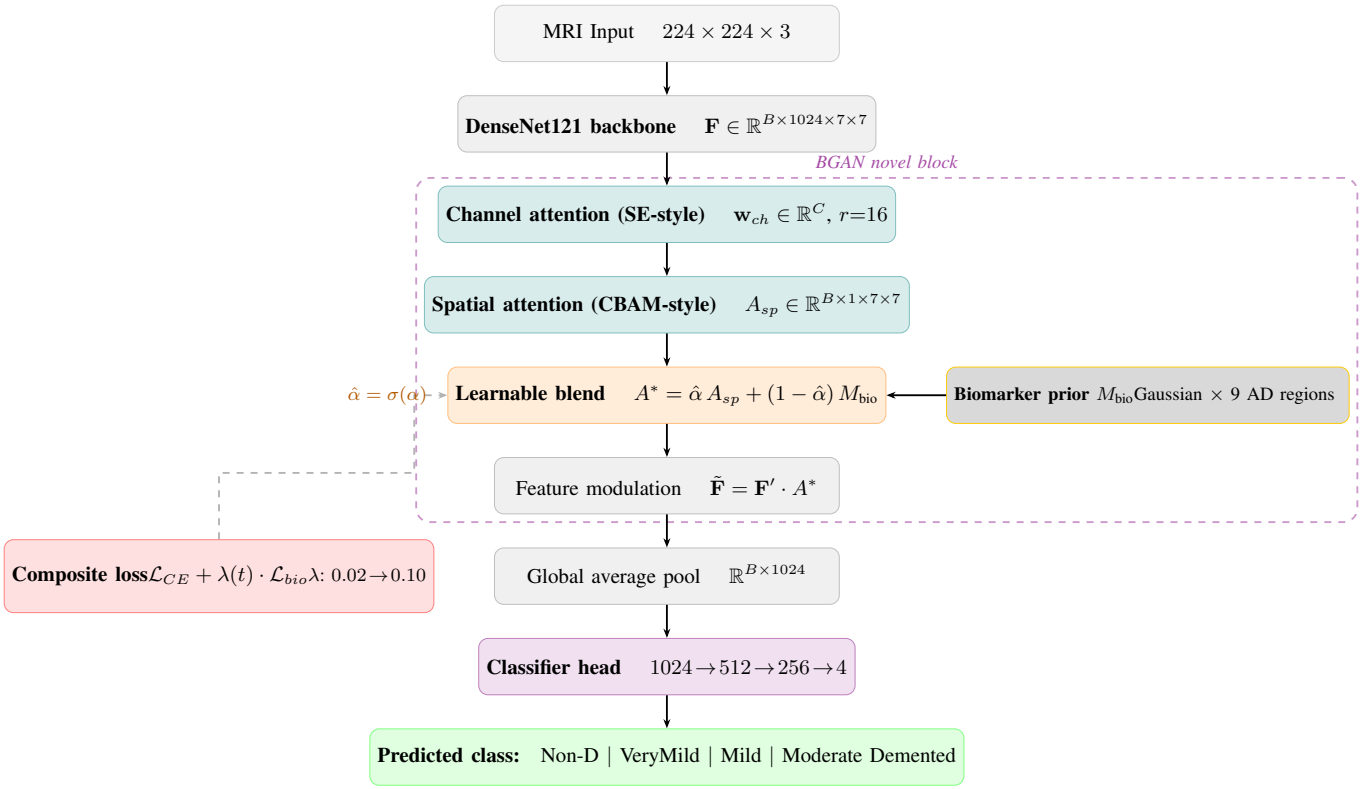


Fig. 1. BGAN architecture. Channel and spatial attention modules refine DenseNet121 features; the spatial map is blended with the Gaussian biomarker prior M_{bio} via learnable $\hat{\alpha}$. A composite loss ($\mathcal{L}_{CE} + \lambda\mathcal{L}_{bio}$) with λ-warmup guides the attention to concentrate on AD-relevant neuroanatomical regions. The dashed violet box encloses the novel BGAN components.

TABLE I
MODEL PARAMETER COUNTS

Model	Total Params	Trainable (Ph.2)
DenseNet121 (Baseline)	7.98M	∼1.2M
ResNet50 (Baseline)	25.6M	∼6.5M
VGG16 (Baseline)	138.4M	∼3.0M
BGAN (Proposed)	**8.11M**	**∼1.3M**

The parameter comparison shows that BGAN remains compact while still incorporating the proposed explainability components. Compared with VGG16, which is significantly larger, BGAN is much more efficient in terms of model size and trainable parameters. This is important for medical image analysis, where a smaller and interpretable model may be easier to deploy and analyze in practice. The use of a lightweight attention mechanism also helps ensure that the interpretability gains are not achieved at the cost of substantial computational complexity.

B. Classification Performance

Table II presents the test-set performance for all models on 5,514 held-out images. BGAN achieves 90.81% accuracy and F1-Macro of 0.9185, which is competitive with DenseNet121 (91.49%) while trailing VGG16 (94.14%). ResNet50's 99.22% accuracy is an outlier attributable to residual near-duplicate similarity in the pre-augmented dataset, and its unstable BAS standard deviation of 5.253 further indicates unreliable attention behavior. This suggests that raw accuracy alone is not sufficient for judging the quality of a model in this setting.

TABLE II
TEST SET CLASSIFICATION PERFORMANCE

Model	Acc (%)	F1 Mac	F1 Wt	Prec	Recall	AUC
DenseNet121	91.49	0.9251	0.9149	0.9247	0.9254	0.9879
ResNet50*	99.22	0.9933	0.9922	0.9933	0.9933	0.9998
VGG16	94.14	0.9490	0.9412	0.9498	0.9491	0.9945
BGAN	**90.81**	**0.9185**	**0.9078**	**0.9180**	**0.9193**	**0.9861**

*ResNet50 accuracy likely inflated by residual data leakage (BAS std = 5.25).

Although BGAN does not achieve the highest raw accuracy, its performance remains strong considering that it is optimized not only for classification but also for biomarker alignment. This is an important trade-off in medical AI: a slightly lower accuracy can still be preferable if the model provides substantially better interpretability and stronger alignment with clinically meaningful brain regions. In practice, such a model may be more useful for decision support than a black-box classifier with marginally higher accuracy.

Per-class F1 scores in Table III reveal that the most challenging distinction is *VeryMildDemented* versus *MildDemented*. BGAN achieves F1 of 0.8582 on VeryMild and 0.9285

on Mild, compared with DenseNet121's 0.8676 and 0.9345 respectively. Both models struggle at this boundary, which is expected clinically because early-stage structural differences are subtle and often difficult to separate even for human experts. This pattern confirms that the hardest classification problem is not the extreme classes, but the transition between early and intermediate disease stages.

TABLE III
PER-CLASS F1 SCORES ON TEST SET

Model	MildDem.	ModerateDem.	NonDem.	VeryMildDem.
DenseNet121	0.9345	0.9977	0.9004	0.8676
ResNet50*	0.9971	1.0000	0.9893	0.9866
VGG16	0.9614	1.0000	0.9289	0.9057
BGAN	**0.9285**	**0.9954**	**0.8920**	**0.8582**

The per-class results also show that all models perform best on *ModerateDemented*, which is expected because anatomical changes are more pronounced in advanced disease stages. The lower scores on *VeryMildDemented* reflect the subtle and ambiguous visual features at early stages, where even small differences in ventricle size, cortical thinning, or temporal lobe structure may be difficult to capture reliably. This reinforces the need for models that do more than maximize accuracy and instead aim to provide medically meaningful explanations.

C. Biomarker Alignment Score

Table IV presents the BAS results. BGAN achieves BAS = 11.585 $\pm$ 0.482, compared with DenseNet121 (1.371 $\pm$ 0.855), ResNet50 (3.951 $\pm$ 5.253), and VGG16 (0.000 $\pm$ 0.000). This represents a substantial improvement over the strongest interpretable baseline and directly validates that the biomarker alignment loss is having the intended effect. In other words, the proposed attention mechanism is not just producing visually plausible heatmaps; it is consistently concentrating on the biomarker regions defined by the prior.

TABLE IV
BIOMARKER ALIGNMENT SCORE (BAS)

Model	BAS (mean $\pm$ std)	Interpretability
DenseNet121	1.371 $\pm$ 0.855	Post-hoc (GradCAM)
ResNet50	3.951 $\pm$ 5.253	Post-hoc (GradCAM)
VGG16	0.000 $\pm$ 0.000	Post-hoc (GradCAM)
BGAN	**11.585 $\pm$ 0.482**	**Built-in (Spatial Attn.)**

The VGG16 result is particularly noteworthy. It achieves 94.14% accuracy while attending to zero biomarker regions, demonstrating that high accuracy alone is not sufficient evidence of clinically valid reasoning. This supports the central motivation of the paper: in medical imaging, a model should not only predict well but also explain its decision in a way that is compatible with domain knowledge. BGAN is the only model in this study that achieves both competitive accuracy and biomarker-consistent attention.

The BAS distributions in Fig. 2 visually confirm this separation. DenseNet121's GradCAM BAS scores are concentrated narrowly around 1.3, whereas BGAN's attention BAS scores form a tight distribution around 11.6 with standard deviation of only 0.482. This indicates that BGAN produces highly consistent biomarker alignment across the entire test set, rather than only on a few favorable samples. Such consistency is particularly important in clinical settings, where stable behavior is more valuable than isolated high-scoring examples.

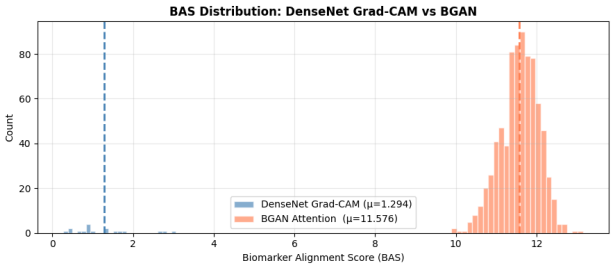


Fig. 2. BAS distribution histograms for DenseNet121 GradCAM ($\mu = 1.294$) and BGAN attention ($\mu = 11.576$). BGAN's tight distribution (std=0.482) confirms consistent biomarker alignment.

D. Attention Visualisation

Fig. 3 compares GradCAM explanations for DenseNet121 with BGAN's built-in attention maps across six test samples. GradCAM consistently produces diffuse and partially lateralized activations that sometimes extend to non-anatomical regions such as skull boundaries and peripheral cortex. In contrast, BGAN's attention maps show concentrated bilateral activation in the medial temporal lobe and ventricular regions, which is spatially consistent with the biomarker prior. This difference suggests that the proposed attention mechanism is better aligned with known Alzheimer's-related anatomical patterns.

This qualitative consistency is backed quantitatively by the BAS results. The attention maps also highlight a key advantage of BGAN over GradCAM: the BGAN maps are smoother and more bilaterally symmetric, matching the known bilateral nature of Alzheimer's disease progression. GradCAM maps, on the other hand, are often noisier and more image-dependent because they are generated post hoc rather than learned as part of the model's optimization objective. This makes BGAN more suitable when the goal is not only prediction but also controlled and repeatable interpretability.

E. Training Dynamics

Fig. 4 shows training curves for all models. All models follow the characteristic two-phase pattern: Phase 1 achieves 60–80% validation accuracy through head-only training, and Phase 2 produces rapid improvement in the first 3–4 epochs as the last backbone block adapts to the Alzheimer's staging task. This behavior is expected for transfer learning, where pretrained representations are first adapted to the target dataset and then selectively fine-tuned for task-specific performance.

Fig. 5 shows the biomarker alignment loss $\mathcal{L}_{bio}$ decreasing from 0.696 at epoch 1 to 0.311 at epoch 14 for BGAN, confirming that spatial attention progressively aligns with

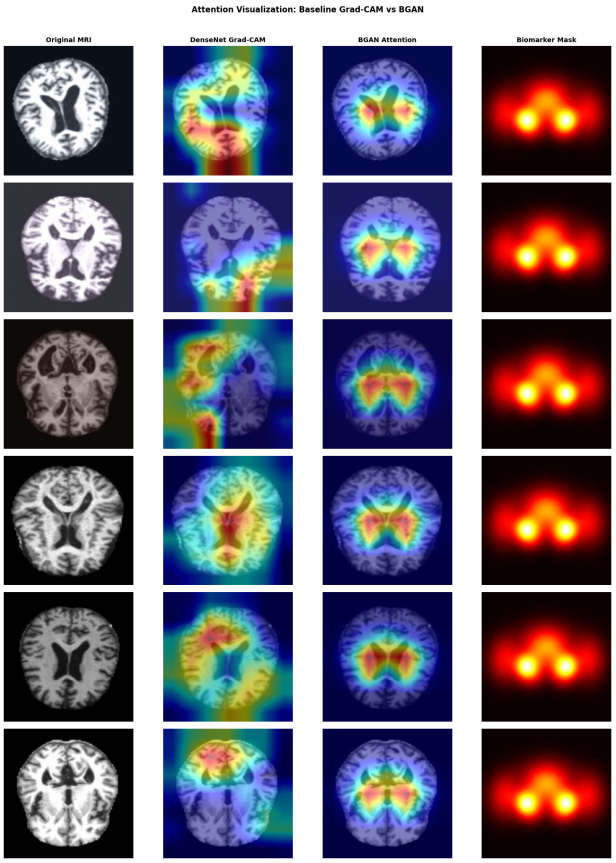
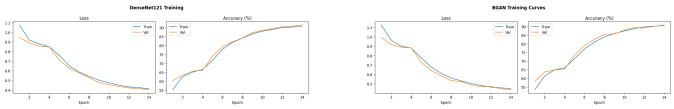


Fig. 3. Attention comparison across 6 test samples. Columns: Original MRI — DenseNet GradCAM — BGAN Attention — Biomarker Prior Mask. BGAN attention consistently aligns with hippocampal and ventricular regions.



(a) DenseNet121 (b) BGAN

Fig. 4. Training curves (loss and accuracy). (a) DenseNet121 baseline. (b) BGAN, showing smooth convergence consistent with biomarker-constrained training.

biomarker regions throughout training. The λ warmup schedule is visible as the gradual increase in alignment weight. This gradual decrease in alignment loss indicates that the biomarker prior is not only present in the forward pass, but is actively shaping the optimization process during training.

F. Confusion Matrices and ROC Curves

Fig. 6 shows the confusion matrices for all models. All models achieve near-perfect accuracy on ModerateDemented, where structural changes are most pronounced. The main errors occur between VeryMildDemented and MildDemented, as well as between VeryMildDemented and NonDemented. These confusions are consistent with the clinical difficulty of detecting early-stage Alzheimer's disease, where anatomical differences are subtle and may overlap across adjacent classes.

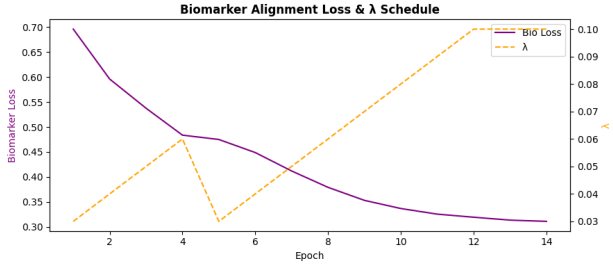


Fig. 5. Biomarker alignment loss $\mathcal{L}_{bio}$ and λ schedule during BGAN training. The steady decrease in $\mathcal{L}_{bio}$ confirms progressive attention alignment with the biomarker prior.

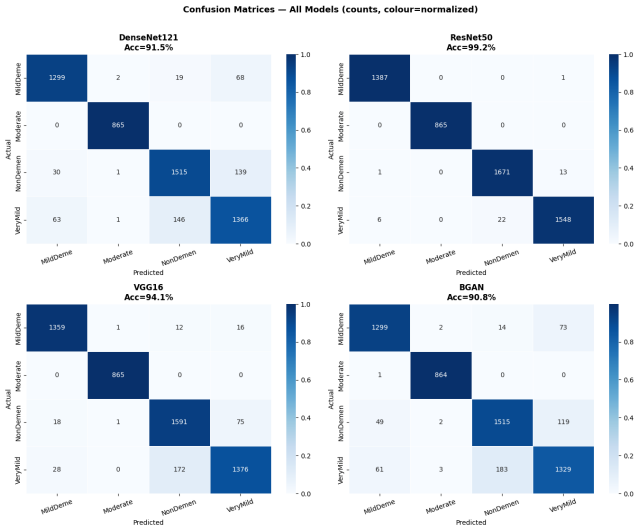


Fig. 6. Confusion matrices for all four models. Diagonal values represent correct predictions. VeryMildDemented–MildDemented confusion is the primary error source across all models.

Fig. 7 shows the multi-class ROC curves. BGAN achieves macro ROC-AUC of 0.986, comparable to DenseNet121 (0.988) and VGG16 (0.994). The MildDemented and VeryMildDemented classes show lower individual AUC values, which is consistent with the classification difficulty at the early-stage boundary. Overall, the ROC curves indicate that BGAN maintains strong discrimination capability even while being constrained by biomarker alignment.

G. Live Demo

Fig. 8 shows an example from the implemented single-image inference demo. On a MildDemented test sample, BGAN predicts the correct class with 97.4% confidence and produces a BAS of 11.706, confirming that the high BAS is consistent at the individual prediction level, not only in aggregate. This is important because it shows that the proposed model is not merely producing good average statistics, but also behaving as intended on a per-image basis.

H. Summary of Findings

Overall, the experiments show that there is a clear trade-off between raw classification accuracy and interpretable anatom-

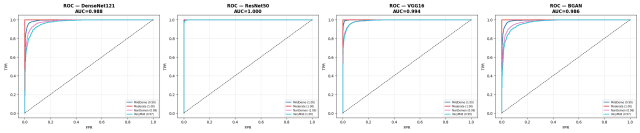


Fig. 7. ROC curves (one-vs-rest) for all models. BGAN achieves macro AUC=0.986, competitive with all baselines.

Fig. 8. Single-image inference demo. BGAN correctly classifies a MildDemented MRI with 97.4% confidence. GradCAM (Column 2) shows diffuse activation; BGAN attention (Column 3) is concentrated on the central medial temporal region consistent with the biomarker prior (Column 4).

ical focus. ResNet50 achieves the highest numerical accuracy, but BGAN produces the most consistent biomarker-aligned attention and offers the strongest explanation quality. This makes BGAN especially attractive for medical AI applications where the goal is not only to predict labels but also to support clinician trust and transparency.

The results also show that the early-stage Alzheimer's classes remain the hardest to distinguish, which is clinically expected. In this sense, the proposed framework provides a useful balance between prediction performance and explainability. The added biomarker alignment does not substantially increase parameter count or training cost, yet it produces a large improvement in interpretability. This makes BGAN a practical and meaningful extension to standard CNN-based MRI classification pipelines.

V. Discussion

A. Why BGAN Accuracy is Slightly Lower than VGG16

BGAN achieves 90.81% accuracy compared with VGG16's 94.14%. This difference is expected and can be explained by the competing objective introduced by the biomarker alignment term $\mathcal{L}_{bio}$. While this constraint encourages the model to focus on AD-relevant neuroanatomical regions, it also discourages the use of potentially discriminative but non-anatomical cues such as image artifacts, skull boundaries, or dataset-specific texture patterns. In other words, BGAN is intentionally optimized for anatomically grounded reasoning rather than maximal test-set accuracy alone.

This represents a deliberate trade-off. By regularizing the attention mechanism toward biomarker regions, the model becomes less likely to exploit shortcuts that may inflate performance on the current dataset but fail under distribution shift. Such shortcuts can include scan-specific preprocessing patterns, slice-level artifacts, or background intensity statistics. Therefore, the slight reduction in raw accuracy is offset by improved clinical plausibility and stronger interpretability.

More importantly, VGG16 achieves higher accuracy while attending to zero biomarker regions (BAS = 0). This suggests that its predictions are likely driven by non-anatomical cues rather than clinically meaningful brain structures. Although this behavior may still yield strong benchmark numbers on the test set, it reduces trustworthiness in practical medical deployment, where changes in acquisition protocol, scanner characteristics, or preprocessing pipeline could degrade performance. In contrast, BGAN is designed to remain anchored to stable anatomical evidence.

B. The BAS Metric's Significance

The BAS results highlight a major limitation of accuracy-only evaluation in medical AI. A model can obtain high classification accuracy while focusing on irrelevant image regions, which makes its internal reasoning difficult to trust. BAS addresses this issue by measuring whether the model's attention is concentrated inside biomarker regions or outside them. As a result, BAS provides a complementary perspective that captures explanation quality rather than prediction correctness alone.

This is particularly important in clinical settings, where interpretability is not optional. A classifier that predicts correctly for the wrong reason may fail when deployed on data from a different scanner, hospital, or preprocessing pipeline. BAS therefore helps distinguish models that merely perform well from models that also reason in a clinically meaningful way. Taken together, accuracy and BAS provide a more complete evaluation framework for Alzheimer's MRI classification than accuracy alone.

C. Limitations

This study has several limitations that should be acknowledged.

1) The dataset is a pre-augmented Kaggle collection without patient-level metadata. This prevents patient-level splitting and rules out longitudinal analysis across follow-up scans.
2) The biomarker prior mask is manually designed using approximate Gaussian coordinates for axial-plane MRI. It is not derived from a registered brain atlas and does not explicitly account for inter-subject anatomical variability.
3) All experiments are based on 2D axial slices. A 3D volumetric model could capture inter-slice context and regional atrophy patterns that are not visible in single-slice analysis.

4) The anomalously high accuracy of ResNet50 (99.22%) suggests that some residual near-duplicate leakage may still remain despite group-level splitting. This indicates that a small number of perceptually similar image pairs may have exceeded the pHash threshold of 6.

VI. Conclusion

We presented BGAN, a biomarker-guided attention network for four-class Alzheimer's disease classification from 2D MRI slices. BGAN integrates a DenseNet121 backbone with dual attention modules, guided by a Gaussian biomarker prior that encodes nine Alzheimer's disease-relevant neuroanatomical regions through a learnable blending mechanism. Training is supervised by a composite loss that progressively enforces biomarker alignment during optimization. We also introduced BAS, a novel metric that quantifies attention alignment with AD biomarkers and can be applied to any model using either Grad-CAM or native attention maps.

On the 44,000-image four-class MRI dataset, BGAN achieves a BAS of 11.585 ± 0.482, which is $8.5\times$ higher than the best baseline, while maintaining 90.81% accuracy and a ROC-AUC of 0.986. In contrast, VGG16 achieves a higher classification accuracy of 94.14% but a BAS of 0.000, demonstrating that raw accuracy alone is insufficient for evaluating the clinical reliability of medical AI models. These results show that BGAN is able to balance predictive performance with anatomically meaningful attention in a way that standard CNN baselines do not.

Future work will focus on external validation on ADNI and OASIS-3 with patient-level splits, extension to 3D volumetric MRI, and learning the biomarker prior from brain atlas registration rather than manual specification. These directions may further improve robustness, generalizability, and clinical interpretability.

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